

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL (style)
ACADEX PRACTICE EXAMINATION
O-LEVEL · COMBINED SCIENCE

COMBINED SCIENCE 5006/2

PAPER 2 June 2024

2 hours

PAPER 2 — STRUCTURED QUESTIONS — ANSWER ALL — 2 HOURS

These are **original ACADEX questions** written to match ZIMSEC syllabus structure, mark allocations and command words. They are **not** leaked or copied official papers. This script is unique to June 2024 5006/2.

Additional materials: Answer all questions. The number of marks is given in brackets [].

INSTRUCTIONS: Write your answers in the spaces provided. You may use a calculator.

- 1.** (a) State one adaptation of a palisade mesophyll cell and link it to photosynthesis. [2]<<
>>(b) State one adaptation of a red blood cell for oxygen transport. [2]<<
>>(c) Describe how a sperm cell is adapted to reach and fertilise an egg. [3]<<
>>(d) Explain why ciliated cells line the trachea. [3] **[10]**
- 2.** (a) Name two components of a balanced diet other than carbohydrate. [2]<<
>>(b) State one function of protein in the body. [2]<<
>>(c) Name the deficiency disease caused by a lack of vitamin C and one caused by a lack of iron. [2]<<
>>(d) Explain why fibre (roughage) is needed in the diet. [4] **[10]**
- 3.** (a) Name three components of blood. [3]<<
>>(b) State the function of red blood cells. [2]<<
>>(c) Describe how a white blood cell can destroy a bacterium. [3]<<
>>(d) Explain why plasma is needed. [2] **[10]**
- 4.** (a) Name the type of pathogen that causes cholera. [1]<<
>>(b) Describe how cholera is transmitted. [2]<<
>>(c) Explain why oral rehydration is used to treat cholera. [3]<<
>>(d) HIV is a virus. Explain how it leads to AIDS, and state one way transmission can be reduced. [4] **[10]**

5. (a) Describe the arrangement and motion of particles in a solid and in a gas. [4] <<
>> (b) Name the change of state from solid to liquid and from gas to liquid. [2] <<
>> (c) Explain, in terms of particles, why a gas can be compressed but a liquid cannot. [4] **[10]**
6. (a) What is an electrolyte? [2] <<
>> (b) Name the products at the cathode and anode when molten lead(II) bromide is electrolysed. [2] <<
>> (c) Explain why solid lead(II) bromide does not conduct electricity but molten lead(II) bromide does. [3] <<
>> (d) State one industrial use of electrolysis. [3] **[10]**
7. (a) Define average speed. [2] <<
>> (b) A learner runs 100 m in 20 s. Calculate the average speed. [3] <<
>> (c) A car travels at constant velocity. What is the resultant force on the car? Explain. [3] <<
>> (d) Name the instrument used to measure time in this experiment. [2] **[10]**
8. (a) State the law of reflection. [2] <<
>> (b) A ray in air hits a glass block. Describe what happens to its speed and direction. [3] <<
>> (c) Explain why sound cannot travel through a vacuum, but light can. [3] <<
>> (d) State how pitch and loudness of a sound relate to the wave. [2] **[10]**

END OF QUESTION PAPER

Worked solutions and mark-scheme notes are inside the ACADEX app (Extract & Study).